

STUDY GUIDE WHO



Agenda 1 : Addressing Challenges in the
Successful Implementation of
Immunization Agenda 2030

Agenda 2 : Developing Strategies to
Combat Infodemics in the Digital Age





Letter from R&D

Dear Delegates,

The Department of Research & Development of MUNSociety MPSTME welcomes you to the 11th edition of Mumbai MUN! Our department upholds a proud tradition of facilitating meaningful discourse on national and international matters, encompassing both historical events and current issues.

We are primarily tasked with formulating agendas and preparing comprehensive study guides. We have created detailed and engaging guides aimed at providing valuable resources for all delegates.

This year's theme, "Redefine the Norm," challenges us to think beyond conventional solutions and embrace innovative approaches to global challenges. In a rapidly evolving world, traditional methods often fall short of addressing contemporary issues. We believe that transformative change requires the courage to question established practices and the creativity to envision alternative pathways.

With this theme in mind, we have thoroughly picked agendas and committees that demand fresh perspectives, challenge conventional thought, and encourage innovative approaches to problem-solving. In this study guide, you will find well-researched background information to help you prepare for your committee sessions.

We, as facilitators, aim to provide you with the intellectual arsenal to combat any challenges that you might face and make the most of your Mumbai MUN experience.

We look forward to your active participation and contributions during Mumbai MUN and are confident that, together, we can make this conference a memorable and enriching experience for all.

Best wishes,

Department of Research & Development,
MUNSociety MPSTME.

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Note: The QARMAs will be released pre-conference, post consultation with the Executive Board.

Agenda 1: Addressing Challenges in the Successful Implementation of Immunization Agenda 2030

COMMITTEE OVERVIEW

The World Health Organisation (WHO) is a specialised agency under the United Nations to deliberate over and improve the standards for public health conditions worldwide.

Established in 1948, WHO signifies its motifs of its broad mandate, by illustrating its primary attainment of “highest level of possible health” by all people irrespective of their country, race, or community.

Based in Geneva, the WHO operates through its general conference via the World Health Assembly (WHA), which accounts as its general policy making body.

The WHO takes charge of assessing and preparing for past, present & future health concerns worldwide, with the aid of its surveillance monitoring systems, in order to control, predict, and prepare for future epidemics. Moreover WHO accounts for running numerous awareness campaigns, sanitary drives in underserved communities, displaced populations, etc. in order to raise standards for healthcare every year. Furthermore, WHO also works with non-government organisations (NGOs) and regional bodies in order to provide resources and incentivise those communities to improve the benchmark of health care goals each year, leading to a healthier world.

INTRODUCTION

An estimated one in five children worldwide still lack access to life-saving vaccines, despite childhood vaccinations preventing nearly four million deaths each year.

Immunisation is a key factor, upholding the infrastructure of Primary Health Care systems (PHCs) worldwide. However, across the globe

there are innumerable underserved countries, and marginalised communities which do not have access to any immunisation services. Vaccines spearhead immunisation, helping combat communicable diseases.

In order to cater to such risks to global health security, the World Health Assembly passed the Immunization Agenda 2030 (IA2030). Immunization Agenda 2030 focuses on its virtues by aiming for over 90% coverage for all vital vaccines, laying emphasis over making Primary Healthcare accessible to all.

BACKGROUND

Goals

The overarching framework of the Immunization Agenda 2030 expands its visions and goals through the following 3 key point:

1. To significantly mitigate the adverse effects pertaining to mortality and morbidity caused by Vaccine-Preventable Diseases.
2. Instigate the research and production for new vaccines and foster an environment with equitable access to all the vaccines to every human, irrespective of their ethnicity, colour, race, tribe, and country.
3. Consolidating Primary Health Care (PHC) worldwide by improving immunisation, effectuating global health coverage, consequently scaling up global sustainable development.

Strategic Priorities

The Strategic Priorities as enumerated by the World Health Organisation, for Immunization Agenda 2030 are thus mentioned below:¹

Immunisation Programmes for Primary Health Care/Universal Health Coverage

Effective, efficient and resilient immunisation services are accessible to everyone as an essential part of primary health care, thereby contributing to universal health coverage.

Commitment & Demand

Immunisation is valued and actively sought by all people. Health authorities commit to ensuring that immunisation is available as a key contributor to the enjoyment of the highest attainable standard of health as a fundamental right

Coverage and Equity

Everyone is protected by full immunisation, regardless of location, age, socioeconomic status or gender-related barriers.

Lifecourse and Integration

All people benefit from recommended immunizations throughout the life course, effectively integrated with other essential health services.

Outbreaks and Emergencies

Immunisation programmes can anticipate, prepare for, detect and rapidly respond to vaccine-preventable and emerging disease outbreaks, and ensure immunisation service

delivery during acute emergencies and among communities affected by conflict, disaster and humanitarian crisis.

Supply & Sustainability

All countries have a reliable supply of appropriate and affordable vaccines of assured quality, and sustainable financing for immunisation programmes.

Research & Innovation

Innovations to increase the reach and impact of immunisation programmes are rapidly made available to all countries and communities.²

Primary Objective

The primary objective for the Immunization Agenda 2030 divulges its versatility to the volatile and sudden needs in the current world with respect to healthcare. By the same token, the Immunization Agenda is different as compared to its predecessor, the Global Vaccine Action Plan (GVAP).

Withal, the Immunization Agenda 2030 (IA2030) articulates its gender responsive strategies, undertaking a fair and equitable interface to both the genders; along with its scheme to ensure the delivery of vaccines to conflict-affected populations. This includes displaced tribes, cross border marginalised communities, and most importantly conflict agitated zones.

IA2030 emphasises over improving healthcare systems overall, consequently improving the sustainability of healthcare systems around the world. This methodology indubitably helps to

¹Data Action Framework for IA 2030. (n.d.-c).

http://www.immunizationagenda2030.org/images/documents/IA2030_Data_Action_Framework_-_FINAL_format_210630.pdf

²A Data Action Framework. (n.d.-b).

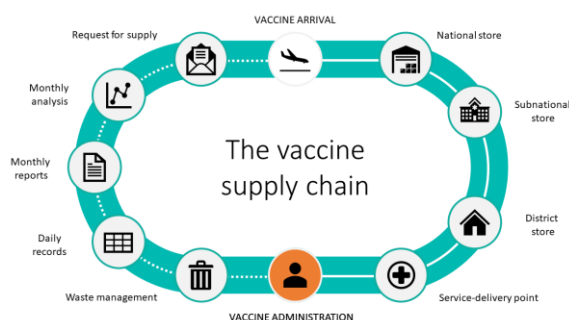
http://www.immunizationagenda2030.org/images/documents/IA2030_Data_Action_Framework_-_FINAL_format_210630.pdf

combat future epidemics, in terms of disease control, vaccine research, and eradication.

Into the bargain, IA2030 accentuates the need for a life-course approach. It suggests that the need for a booster dose, as one grows into adulthood, must be duly administered in order to improve the immunity overall for each human worldwide. Besides healthcare, IA2030 believes deeply in collaboration. It contemplates and portrays the integral need for partnerships and new alliances, in order to execute the common goal for world health, not limiting the cause to the agenda's tenure.

CHALLENGES

Logistics



The Vaccine Supply Chain

Source: WHO

The delivery of vaccines is a lengthy procedure. Once the monthly reports and statistical data is analysed, there is a request for vaccines in a given location. Once the vaccines are allotted to their respective regions, they are sent to national stores, which further transport the vaccines to sub-national stores. Lastly these vaccines are then shared with NGOs, hospitals, and service delivery points to administer the doses.³

Verily, Immunisation programmes, or any major healthcare efforts, are carried out with the aid of a strong and resilient supply-chain system, which assert to be the backbone of the programme. The handling of vaccines, equitable and fair distribution of vaccines depends on the logistical support management. The Immunization Agenda 2030, is an effort with its tenure lasting a decade, urges itself to evolve its pre-existing cold chains. Undoubtedly, on-ground management is necessary to administer a fair and equal share of vaccine supply to all underserved communities and thus keep the practice clean, and free from any political/regional financial agitation. Problems caused in logistics often include correctional costs, loss of resources, miscommunication and various other adverse effects. These problems hamper the efficiency of the logistic supply chain and thus create challenges in the longer run. Furthermore, the need to invest in a good logistical infrastructure attests to a good management and delivery of vaccines, as well as the effective disposal of the vaccines, in order to reduce its environmental footprint regarding the same.

Finances & Resources

Expenditure indicators from the pre-COVID era illustrate the average domestic spendings among low and lower-middle class incomes to be very constrained.

In 2019, the average public spending on health was \$9 per capita(p.c.) in lower income countries, \$49(p.c.) in lower-middle income countries, \$255 in upper-middle income countries, and lastly \$2,234 in high-income countries. The difference in the share of overall health expenditure, from 29% to 12% in lower-middle income and low-income countries, paints an important indicator

³World Health Organization. (n.d.). *Logistics & Supply Chain Management*. World Health Organization. <https://www.who.int/teams/immunization-vaccines-and-biologicals/essential-programme-on-immunization/supply-chain>

of public health spending. This also signified the different trends in the costs of vaccines in each country. Costs of vaccines prospectively depend upon the age distribution in the population, coverage levels, prices paid, along with the costs incurred by the countries. About nearly half of both PHC and non-PHC spending in low and middle income countries was funded by NGOs such as the GAVI foundation and the Bill & Mellinda Gates foundation, aiding these countries to overcome financial roadblocks in the country's healthcare development.

One of the most integral aspects while working towards immunisation includes domestic financing. Development partners and Civil Society Organisations (CSOs) thus streamline their roadmap and intentions as per the constituted strategic priorities as laid out by the Immunization Agenda 2030.

This approach demonstrates a more transparent preface to their contributions with respect to IA2030's goal. Additionally, it helps render accountability for IA 2030's goals, crediting to the better data analytics and monitoring systems to assess the trends for the same.

Technological adaptation

The Immunization Agenda 2030 aims for fair and equitable access to vaccines across the globe. Technological adaptation stands out as a major barrier in this regard. The most recent vaccine technologies such as innovative delivery systems, thermostable formulations, and integrated devices find difficulty in being adopted on a large scale. For this reason, the vaccine industry has been considered risk-averse in character since the introduction of new technologies usually involves heavy capital investment, long clinical trials, and regulatory clearances.

There is also a gap between innovation in technology and market demand. While producers of vaccines often are moving forward on new technologies, purchasers and policy makers who buy vaccines and make vaccine policies often lack awareness or incentive to incorporate new technologies. For example, promising approaches such as microneedle patches or dual-chamber syringes may experience delays in being put into practice due to a procurement system that focuses primarily on the lowest cost rather than the benefit of innovation. Such a gap makes improved immunisation tools inaccessible in a timely manner for those most in need in low and middle income countries, who already face difficulties with current infrastructure.⁴

Public Perception

The Immunization Agenda 2030 looks forward to developing a very robust and omnipotent Primary Healthcare structure. Initialization of circulating a new vaccine often requires public and communal support for the idea to pass through. In addition, the introduction of any new medication results in innumerable false misconceptions, which hamper public confidence in accordance with embracing the new vaccines and medications.

Thus, the use of mass media, or on ground local media is often important in order to rightfully spread awareness regarding the need for immunisation.

In addition, public perception in marginalised communities, secluded tribes, remote villages, etc. is vital for Immunization Agenda 2030 because it lays emphasis over making healthcare and vaccination to be accessible to all; and especially to raise awareness in such underserved settings.

⁴ D., K. D. (n.d.). *Strategies to advance vaccine technologies for Resource-poor settings*. Vaccine. <https://pubmed.ncbi.nlm.nih.gov/23598477/>

PAST INTERNATIONAL ACTION

GVAP Polio Endgame Strategy (2019-2023)

Admittedly, on the premise of GPEI's (Global Polio Eradication Initiative) success which reduced World Poliovirus (WPV) drastically by 90% worldwide, WHA launched the GVAP Polio Endgame Strategy to eradicate WPV globally in its entirety. Essentially, this plan enkindles its frameworks to address 3 main key points, as signified below:

1. Insecurity & conflict
2. Weak or fragile health systems
3. Operational, Management & resource risks.

Additionally, the GVAP plan asserted 2 major goals, which namely are;

1. Eradication of WPV with regards to Vaccine-Derived Polio (cVDPV) Outbreaks and eliminate any mere possibilities of emergence of WPV outbreaks in the foreseeable future.
2. Integration
 - Constituting a framework to strengthen health systems worldwide in order to consolidate immunisation.
 - Improving surveillance systems to enforce a comprehensive approach to future epidemics, vaccine-preventable diseases (VPD) & communicable diseases, with the aid of data analytics, credited to the surveillance systems.
 - To improve logistics and thus prepare for an immediate response for any future epidemics.

- Certifying a containment and a successful eradication of World Poliovirus (WPV) globally.

GVAP, Global Measles & Rubella Strategic Plan (2012-2020)

Aimed to achieve and maintain immunisation against measles, rubella, and congenital rubella syndrome (CRS); the GVAP reduced mortality rate by 95% as compared to the year 2000 over its tenure. The GVAP plan's omnipotence was portrayed by its 5 main key features, namely:

1. A relatively high double-dose vaccine coverage, consequently helping us attain high levels of immunity worldwide.
2. Ensuring growth, with moderated surveillance and use of data analytics to track past records.
3. Developing a strong healthcare infrastructure to improve responses to mitigate any imminent or future epidemics.
4. Optimal use of mass media to cultivate awareness, along with the need for immunisation.
5. Engaging in research and development, consequently improving standards for diagnostic tools, and thus aiding a rather cost-effective approach to on ground operations.

The GVAP Global Measles & Rubella Strategic plan signifies an archetypal framework which indubitably complements all the Measles and Rubella mitigatory efforts, which include the following: country ownership, strengthening regulatory immunisation, ensuring and certifying camaraderie with other health interventions, and lastly ensuring healthcare to be accessible by all.



Measles, Mumps and Rubella vaccine
Source: Times Now

South-East Asia Regional Vaccine Action Plan 2030

The synchrony of IA 2030, along with the framework for South-East Asia Regional Vaccine Action Plan 2030, outlines the primary objective of addressing regional priorities. Inevitably, countries in the region faced challenges with the COVID-19 pandemic, which urged the assembly to renew the plan after the original RVAP(2016-2020); which indeed did bolster the immunisation levels across the region over the course of its tenure. Essentially it underlines the significance of regional monitoring systems, to further evaluate the norms deliberated by the Seventy fourth World Assembly. Moreover, this approach succours it to supervise an effective execution of Immunization Agenda 2030, which would be thus covered in the course of 2022-2026⁵.

European Vaccine Action Plan (2015-2020)

The European Vaccine Action Plan was constituted to cater to challenges regarding immunisation in the European regions. The

action plan was streamlined to control all Vaccine Preventable Diseases (VPDs) in its entirety. Furthermore, the plan deliberated over the pre-existing regional healthcare policies & Health 2020. Health 2020 is a European policy framework which looks forward to improving the status of health, eradicating bias amongst individuals with respect to healthcare, through a combined administrative effort and better governance. The European Vaccine Action Plan signifies itself over 6 primary goals listed as follows:

1. Certifying Polio-free status.
2. Control Measles and Rubella.
3. Control Hepatitis B.
4. Address regional vaccine coverage at all administrative levels.
5. Utilising past data and surveillance systems to introduce newer vaccines.
6. Achieving financial sustainability in order to equitably support all national immunisation and healthcare programmes.⁶

KEY PARTIES INVOLVED

International Bodies

WHA (World Health Assembly)

Held annually each year around May, the World Health Assembly holds power to exercise decisions passed by WHO (World Health Organisation). The World Health Assembly advocates reviewing the regulations & policies constituted by the WHO. Moreover, WHA

⁵World Health Organization. (1970d, January 1). *Strategic framework for the south-East Asia Regional Vaccine action plan 2022-2030 as aligned with the Global Immunization Agenda 2030*. World Health Organization. <https://iris.who.int/handle/10665/343756>

⁶World Health Organization. (1970b, January 1). *European Vaccine Action Plan 2015-2020*. World Health Organization. <https://iris.who.int/handle/10665/340400>

overlooks the financial policies, and attests to review and approve the proposed programme.

The World Health Assembly, with its partners and member nations thereby voiced the Immunization Agenda 2030, aiming to save more than 50 million lives.

With the COVID 19 pandemic, the WHA aims to improve health and healthcare systems to mitigate any future epidemics, envisioning a healthier future for all.

WHO (World Health Organisation)

Irrefutably, the World Health Organisation plays an integral role in setting the scope for healthcare development in all the countries. It evinces and engenders reasonable agendas, which are further considered, discussed and exercised by the WHA. Additionally, WHO accounts for aiding on-ground support to countries, and sculpting the standards for the health and evidence based policies. WHO being the largest international organisation in terms of healthcare, has effectuated the Immunization Agenda 2030 in its wholeness. Reflecting back upon past epidemics and learning from them, the World Health Organisation strives to maintain high immunisation levels. Furthermore, it believes that its well-constituted approach towards improving healthcare, the IA 2030, is the best investment money can buy." This highly regards the Immunization Agenda 2030 to be one of the greatest health care policies constituted till date in terms of global healthcare.⁷

UNICEF (United Nations Children's Fund)

UNICEF's mandate describes its efficiency, adjudging to its credible work done in the course of the past 75 years.⁸

UNICEF derives its mandate by the comprehensive approach to immunisation as portrayed by the Convention on the Rights of the Child. It reinforces the statement that no child should be deprived of any health care access, and should be provided with the appropriate and necessary healthcare measures, and to be treated fairly at all given instances.

In addition, as a leading advocate in terms of child welfare and healthcare, UNICEF urges its partner bodies to help countries and bolster their healthcare infrastructure in order to make vaccines and healthcare relatively more equitable and accessible.

Regional organisations

European Immunization Agenda 2030(EIA 2030)

Portraying a collective and united outbreak response, addressing clefts in vaccine and immunity coverage amicably among 53 member states, the European Immunization Agenda 2030 speaks volumes for its cooperative efforts to participate with other regional, sub-regional and institutional sectors. EIA 2030 has been essential for highlighting disease control, by mitigating the presence of Vaccine Preventable Diseases (VPD). Furthermore, EIA2030 strives to improve equity and accessibility for vaccines across its regions, and also vitalize its Primary Healthcare in its underserved populations.⁹

⁷Implementing the immunization agenda 2030. (n.d.-c).

https://www.immunizationagenda2030.org/images/documents/IA2030_FrameworkForActionv04.pdf

⁸UNICEF immunization roadmap to 2030. (n.d.-d).

<https://www.unicef.org/media/138976/file/UNICEF%20Immunization%20Roadmap%20To%202030.pdf>

⁹World Health Organization. (n.d.-a). *European Immunization Agenda 2030*. World Health Organization.

<https://iris.who.int/bitstream/handle/10665/250673/9789241511551-eng.pdf.txt?sequence=3>

African Union

The African Union (AU), in unison with the Gavi, the Vaccine Alliance stand together in order to boost the immunisation levels in Africa. This agreement upscales the number of vaccines provided, adding to their arsenal to bolster the immunisation goals in Africa, strengthening their healthcare systems. This agreement was an Memorandum of Understanding (MoU), signed on the behalf of the chairperson of the African Union, and the AU commissioner of Health, striving to reach one prime goal - for everyone in Africa, irrespective of where they live, should have access to immunisation. Furthermore, the African Union Commission (AUC) and Africa centres for Disease Control & Prevention (Africa CDC) launched a Public Health Order, which set the roadmap for primary health care goals for Africa, streamlining itself with the virtues of Immunization Agenda 2030.¹⁰

Private Sectors

Gavi, the Vaccine Alliance

Over the course of the past 2 decades, Gavi has succeeded to vaccinate and immunise over 1 billion children, and halve the mortality rate amongst 78 underserved countries.¹¹

It sustains and builds the omnipotent camaraderie of donor governments, WHO, UNICEF, the World Bank, US centres for Disease Control & Prevention (CDC), Vaccine industries and other technical agencies; in order to bolster

its production and execution of life saving vaccines worldwide.

Gavi boasts a good stockpile management, and on-ground logistics, crediting its partnerships worldwide. Additionally, its strength to share expertise and research, it stands to maximise the effectiveness of administering new vaccines, and delivery of vaccines.



Collaborative Partners Driving Gavi's Mission

Source: GAVI

Bill & Melinda Gates Foundation

Partnering with countries and multilateral organisations, civil-society organisations, and private sectors, Bill & Melinda Gates foundation signifies the addressed need for strong PHC systems worldwide. This includes proper sanitation, implementation of all the healthcare policies; to provide a structure to build a stronger healthcare ecosystem altogether.¹²

Moreover, the Bill & Melinda Gates Foundation caters to the financial needs for underserved

¹⁰Signing of a new agreement to drive vaccine impact in Africa. Signing of a New Agreement to Drive Vaccine Impact in Africa | African Union. (2024, November 1). <https://au.int/en/pressreleases/20230515/signing-new-agreement-drive-vaccine-impact-africa>

¹¹Gavi, the Vaccine Alliance. Global Health Progress. (2024, June 18). <https://globalhealthprogress.org/collaboration/gavi-the-vaccine-alliance/>

¹²Bill & Melinda Gates Foundation. (n.d.). Bill & Melinda Gates foundation and IA203. Bill & Melinda Gates Foundation. <https://www.gatesfoundation.org/our-work/programs/global-development/immunization>

populations, to justify and provide a rather more equitable delivery of high quality vaccines. The access to vaccines is not only limited to the bounds for children & their families, but is consequently quantified to help entire communities and economies. Assertively, the Bill & Melinda Gates Foundation also funds and closely works in conjunction with the Gavi, in order to bolster the reach of vaccines and increase affordability and the mobilisation of all the vaccines.

CASE STUDIES

Polio Virus

Poliomyelitis (Polio) is a communicable disease that majorly affects children under the age of 5. Spread virally through contact, faecal/oral route, Polio multiplies in the intestine, wherein it can encroach the nervous system and consequently cause paralysis. Unfortunately, Polio stands to be one of the diseases without any direct apparent cure, and could only be treated by immunisation. The vaccine for polio is administered over the course of childhood to adolescence in multiple doses, aiming to alleviate the symptoms for Polio.

Launched in 1988, the Global Polio Eradication Initiative (GPEI) is credited for its honorary work to have successfully mitigated about 1.5 million deaths, with the help of its health care infrastructure worldwide. Regardless, accounting to on-ground situations in Afghanistan & Pakistan aggravate the current conditions in immunisation with respect to Polio. This surge in wild Poliovirus in these regions can evince a resurgence and hamper all the past immunisation efforts. In a given scenario, failure to contain Poliovirus in these areas could cause about 200,000 cases every year over the course of

the coming decade. The global effort to eradicate Polio has been declared as a Public Health Initiative of International Concern. Ever since 1994, the WHO region of Americas was declared as a Polio-free region, subsequently followed by the WHO Western-Pacific region and the European region in 2000 and 2002 respectively. In March of 2014, the WHO South-East region spanning over 11 countries from India to Indonesia was certified Polio-free. Lastly in 2020, Africa was one of the regions to be certified Polio-free, crediting to WHO's valiant efforts along with its GPEI partners.

Measles & Rubella

Despite the availability of a direct cure for these, Measles & Rubella is one of the prime reasons for mortality amongst young children. The virus is one of the leading causes of deaths in infants and adolescents, especially in the terms of Vaccine Preventable Diseases (VPDs). Initially, with the use of surveillance systems in 2005, Measles & Rubella averaged about 20 cases per week, which gradually plummeted to about 4 per week in about a decade, in 2015.¹³ In 2017, the Regional committee constituted the Regional Strategy & Plan of Action for Measles and Rubella elimination in the Western Pacific, subsequently demanding each member state to eliminate Rubella with a set of target dates. 8 countries have been certified for Measles elimination and 7 for Rubella. When a pregnant woman is infected with Rubella, it holds a 90% chance that it would get to the foetus and may cause congenital rubella syndrome.

Admittedly, there is no antiviral treatment for Measles and Rubella, and can only be cured by vaccination, for example with vaccines such as "measles-rubella (MR)", "measles-mumps-rubella-varicella (MMRV)", etc. which are often

¹³Polio Virus Centers for Disease Control and Prevention. (2022, December 15). *Progress toward measles and rubella elimination*. Centers for Disease Control and Prevention. <https://www.cdc.gov/mmwr/volumes/71/wr/mm7150a1.htm#:~:text=Measlesandrubellaoutbreak-basedsurveillancestartedin2005with,fivecasesper4weeks.>

administered over the course of 2 doses. In the light of the past few years since 2022, there has been a surge by 255% in the number of Measles cases in the Western-Pacific region, significantly suggesting the harm caused due to the agitation in the movement in the displaced populations due to the COVID 19 pandemic.¹⁴

CONCLUSION

To conclude, the Immunization Agenda 2030 has brought forth a much needed and robust far reaching Immunisation framework to support global health. It focuses on offering widespread immunity and supporting healthier living. It aims

for universal vaccine coverage and should eventually reach every individual regardless of their origin, nationality or background, ensuring equitable vaccine access in underserved communities. By strengthening logistical networks, forging partnerships with regional, international, and non-governmental organisations, it sets the standard of the future for health policy. IA2030, in its own unique ways, represents the world promise of improved health care all over the globe with an intention to overcome vaccine issues. Through this, the WHO affirms its role of leadership by continuously adapting and enhancing healthcare norms to meet evolving global needs.

¹⁴World Health Organization. (n.d.). *Measles and rubella WPRO*. Measles & Rubella - World Health Organization. https://www.who.int/westernpacific/health-topics/measles#tab=tab_3

Agenda 2: Developing Strategies to Combat Infodemics in the Digital Age

INTRODUCTION

We live in a digital age where social networks and other media allow us to share information readily. However, this quick and unverified sharing of information can lead to something called an “infodemic”. The World Health Organization (WHO) uses the term infodemic to describe an overabundance of information – some accurate and some not so much – in times of crisis, making it difficult for people to get access or guidance from reliable sources.

The COVID-19 pandemic proved to be extreme, and the severity of infodemic spread amidst the pandemic reflected the same. There were outspread determinants of health issues to the masses such as conspiracy theories, deception, and news hoaxes thus instilling distrust in the institutes. With the growing use of social media and citizens reporting news, the situation started worsening because false information was being exaggerated and captivating, and had the potential to be dispersed more rapidly than the truth.

To address the spread of infodemics, targeted measures should be undertaken in several areas, including policy development, media and information literacy, technology, and collaboration at the global level. Well-designed approaches directed towards this problem are important not just for public health but to also make more knowledgeable and well-informed choices.

BACKGROUND

According to the WHO, An Infodemic is defined as, “An inordinate amount of information – some accurate and some not so accurate - that is present during an epidemic”. The dissemination

of information that is false or even inclining towards rumours could be calamitous to public healthcare efforts as it could promote needless doubt, suspicion and even detrimental decision making.

The term “Infodemic” is a combination of “information” and “epidemic”, which highlighted how misinformation could spread like a disease. It was during a gathering of a foreign policy and security experts in Munich that the Director-General of the World Health Organization stated “We’re not just fighting an epidemic; we’re fighting an infodemic” referring to how misinformation could be a potential threat.

They pose significant challenges to public health and safety. Misleading information could also influence behaviours, such as vaccine hesitancy, ultimately jeopardising public health efforts and, incorrect claims could result in individuals forgoing effective medical advice. Additionally, infodemics can overpower highly reliable sources, further making it very difficult for people to differentiate between accurate and incorrect information, hence making it immensely complicated in critical situations. This is also seen when common social networking applications further escalate the problem by facilitating rapid information sharing, which is not inherently negative, but can quickly disseminate misinformation.

Amplification

Misinformation proliferation refers to the ease with which weak and inaccurate information is communicated and disseminated, especially by using digital platforms such as social media. This could be done accidentally but the repercussions may be unfavourable especially when such information determines or changes the opinions,

decisions and actions of individuals or groups. Misinformation proliferation has various aspects that make it a notable challenge to consider.

According to a 2018 MIT study, fake news travels six times as quickly as true stories on Twitter.¹⁵ In the first months of the pandemic, misinformation about COVID-19 reached billions of people around the world. For example, a study from the ¹⁶Reuters Institute for the Study of Journalism found that misinformation claims collected from January to March 2020, for a sample, had been seen by at least 1 billion people in 25 countries by the end of June. Another aspect is amplification. The more people interact with the lie, the farther the lie gets to reach based on increasing engagement. For instance, the Facebook-owned platform Instagram has been shown to boost misinformation posts about COVID-19 vaccines because such content often garners more likes and shares. A study found that 37% of COVID-19 misinformation on Twitter was linked to bots, which automated the spread and amplified false information. Lastly, information distortion also plays a role. With every lie that is told, and re-told about a certain issue, the original basis becomes far and far away as more lies begin to take shape.¹⁷ A study of COVID-19 misinformation by First Draft News, an organisation brought together by the Google News Lab, found that 59% of misinformation began with some factual element, such as a

genuine statement from health officials that was later distorted to support false narratives. This creates an added dimension to the challenge of identifying misinformation, as the false claim could have had a somewhat factual foundation which was distorted.

Some examples were false claims such as “COVID-19 vaccines cause infertility”, that have spread wide across the media wherever social support is present, even though it has been debunked by facts and scientists. Such misinformation can lower the dispensing of vaccinations¹⁸ to the population ideally affecting the health of the community. Another example is the well-known health hoax that vaccines cause autism. This false claim originated from a now-debunked study by Andrew Wakefield in 1998, which suggested a link between the MMR (Measles, Mumps, and Rubella) vaccine and autism. Extensive research, including studies from the US centres for Disease Control & Prevention (CDC) and the WHO, has found no link between vaccines and autism. For example, a 2019 study involving over 650,000 children in Denmark found no increased risk of autism after vaccination.

A very powerful tool in the context of misinformation is emotional appeal, which

¹⁵Peter Dizikes | MIT News Office. (n.d.). *Study: On Twitter, false news travels faster than true stories*. MIT News | Massachusetts Institute of Technology. <https://news.mit.edu/2018/study-twitter-false-news-travels-faster-true-stories-0308>

¹⁶Ferreira Caceres, M. M., Sosa, J. P., Lawrence, J. A., Sestacovschi, C., Tidd-Johnson, A., Rasool, M. H. U., Gadamidi, V. K., Ozair, S., Pandav, K., Cuevas-Lou, C., Parrish, M., Rodriguez, I., & Fernandez, J. P. (2022, January 12). *The impact of misinformation on the COVID-19 pandemic*. AIMS public health. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9114791/>

¹⁷World Health Organization. (n.d.-a). *Immunising the public against misinformation*. World Health Organization. <https://www.who.int/news-room/feature-stories/detail/immunizing-the-public-against-misinformation>

¹⁸Beusekom, M. V. (2024, June 1). *Study: Truthful yet misleading Facebook posts drove COVID vaccine reluctance much more than outright lies did*. CIDRAP. <https://www.cidrap.umn.edu/covid-19/study-truthful-yet-misleading-facebook-posts-drove-covid-vaccine-reluctance-much-more>

includes psychological factors worth considering such as cognitive biases and social dynamics.

We often see people are predisposed to favour information which usually confirms their existing beliefs, a phenomenon commonly referred to as confirmation bias. Emotionally charged misinformation which aligns with these beliefs are potentially more readily accepted by the public. Looking at confirmation bias, it is the tendency to seek out and interpret information that confirms pre-existing beliefs while ignoring contradictory evidence. Its impact could potentially involve individuals sharing or believing information which aligns with their views and perspectives, making it more likely to spread within like-minded communities. A similar observation could simply be the tendency of individuals to adapt to certain beliefs and behaviours simply because others are doing the same, this is certainly the “bandwagon effect”.¹⁹

Exploitation of Misinformation

Moreover, there are various reasons why some parts of society may intentionally spread health misinformation during a disease outbreak. Health-related fears have turned into tools for political and economic manipulation, used during the time of elections and even with some as a basis of scams. In health crises, local authorities can exploit what is feared by the populace in terms of diseases or vaccines to push their respective political agendas, citing concern for public well-being for such an excuse. As these health-related fears continue to hold the limelight, media houses increasingly take a role in the diffusion of health concerns, thus inadvertently amplifying infodemics. Scams also sell fraudulent vaccines, cures, or supplements

based on the kind of health fears spread amongst people, especially during times of pandemics. Such schemes often permeate social media, using fake endorsements or posing as medical professionals to lend credibility to unproven products. Such tactics may yield profits for those involved but pose serious risks to public health and exploit economically vulnerable populations.

The pervasive fear of virus-related health, such as COVID-19, has been exploited by the politicians and scammers for their personal gain and leverage. In the context of a health crisis, disputes about vaccine development and disease control are often political as they provoke polarising opinions, strong emotions and disputes in society. For instance, during the US 2020 presidential elections it was possible to uplift the hydroxychloroquine propaganda for some political figures to undermine the public confidence in governmental policies regarding COVID-19 control. Thus, not only did this breed a lack of confidence in health care givers, but also in governmental institutions, and this had an effect on both civic trust and voting patterns.²⁰



Vaccine mandate protests in Los Angeles

Source: LA Daily News via Getty Images

In the case of vaccine safety, for instance, the conception of targeting health fears regarding vaccine safety was used. The promotion of these

¹⁹ Confirmation bias. Confirmation Bias - an overview | ScienceDirect Topics. (n.d.).
<https://www.sciencedirect.com/topics/psychology/confirmation-bias>

²⁰ The emergence of covid-19 in the U.S.: A public health and political communication crisis | request PDF. (n.d.-b).
https://www.researchgate.net/publication/341717979_The_Emergence_of_COVID-19_in_the_US_A_Public_Health_and_Political_Communication_Crisis

conspiracy theorists on social media created an environment where all sorts of unqualified narratives compelled some people to have more fear of a vaccine than the disease, leading to social division even in the domestic and community spheres.

Fraudulent entities abuse the infodemic by spreading false health information with the intention of scaring or providing false security to customers so that they may use their services and products. During disease outbreaks, they take advantage of this uncertainty by peddling disastrous 'wonder drugs', unproven preventive drugs and unsubstantiated herbal products. Utilising platforms with large audiences like Facebook, TikTok, and other online marketplaces, they often run targeted advertisements to lure people seeking health solutions by using emotional words and imagery. Some even masquerade as health experts or organisations to remotely endorse their claims, making it difficult for the public to differentiate fake from real. Such techniques also not only breach the public from real health advice, but also result in the loss of money and rise in health hazards to those who get trapped in these schemes. These scams however can be countered by government health agencies on a continuous basis, although the constant change in the means and methods of the con artists becomes a barrier.

As diseases track the interests of the audience and encourage them to consume content genres, media platforms regularly spread health concerns even if they are infodemics. This strategy exploits the audience's interest, especially in extreme cases such as the COVID-19 pandemic, where the relentless assiduous coverage allows for high viewership and click rates. Exaggerated risks, false claims of cures, and ominous predictions of the future do not allow for substantiated accounts to be released

because people are inclined to disseminate information that arouses emotions. This leads to a situation where media indecisively binds themselves to what is currently the most popular but can be misleading and fill the audience with unnecessary excitement. In this context, the algorithms of social networks contribute to even greater dissemination of "viral" health news, both true and false.

CHALLENGES

In the battle against micro-posts posted on a multitude of entirely different private digital platforms, all managed by specific algorithms, which often encourage sensational content and prioritisation, conveying and amplifying health misinformation is key. This kind of unstoppable flow and pace is fatal. The global digital information overload paired with cultural behaviour within different societies right now means that no health message can be of the type "one size fits all".²¹ The WHO must also navigate complicated partnerships with social media platforms to respect the human's right to freedom of speech and counteract misinformation that has a negative impact on health. In addition, scarce manpower and financial resources continue to impede WHO's capacity of delivering authoritative health information in diverse languages and regions leading to an inequitable access for the indexation of scientifically validated knowledge.

In the future, sophisticated misinformation is likely to pose an even greater threat for WHO and may well involve content created by AI or deep fakes that can make false information look very convincing. This will demand higher capabilities of monitoring and verification technology with the requirement for enhanced financial provisions as well technical knowledge. With

²¹Trachtenberg, A., & Wt. (2024a, February 9). *Pov: Health misinformation is rampant on social media*. Boston University. <https://www.bu.edu/articles/2024/health-misinformation-rampant-on-social-media/>

new health threats and channels for spreading misinformation continuing to present themselves, WHO will have to be agile in response, probably focusing even more on big data together with artificial intelligence analysis of that real-time information. Increasing worldwide digital health literacy and building public trust in official healthcare information sources will be just as crucial, but equally challenging to achieve amidst the misinformation erosion happening right now.

Offline misinformation is likely to be circulated through word of mouth in remote areas where it is difficult to get a new network of information. In such areas, people try to ignore information that is gained from the media and the internet and look into trusted individuals within the community. Family members, friends or influential leaders are such trusted sources. This trust leads to the spreading of false information witnessed in social events like markets or religious gatherings where incorrect information can be shared. In rural settings, the inaccessibility to fact checks makes populations gullible towards any hearsay that usually results in some hysteria and defiance to the public health measures. The WHO points out that dealing with offline misinformation is a major concern as it is also one of the barriers to vaccination uptake, increases the chances of hesitancy and causes harm due to promoting dangerous alternative health behaviours.²²

KEY PAST INTERNATIONAL ACTIONS

Collaboration with Technology Companies

The WHO works with large technology companies such as Google, Facebook, Twitter and YouTube to combat the diffusion of fake information. These platforms also use features such as placing prominent labels which state fact-checking, modification and placing pop-up notifications that redirect users to related verified sources.²³

Sites are encouraged to modify the settings that direct users to post content likely to be false or provocative. Algorithms used by social media can therefore be changed to filter misleading health information thus containing the infodemics.²⁴ Under health categories of videos, the WHO provides fact-sheets along with similar health authorities, and the WHO created its own systematic programs to deal with infodemics, by flagging content.

The WHO approach to health misinformation is tagging false content for fact-checking, removal of flagged health misinformation across platforms, and banning inaccurate content. Healthy behaviour recommendations from the WHO are incorporated into Google and YouTube's global public health campaigns, where misinformation tracking allows the WHO to intervene with harmful messages in real time.

²² IJNet Contributor Muskan Bansal These initiatives leverage local communities to fight false information in the US. International Journalists' Network. <https://ijnet.org/en/story/these-initiatives-leverage-local-communities-fight-false-information-us>

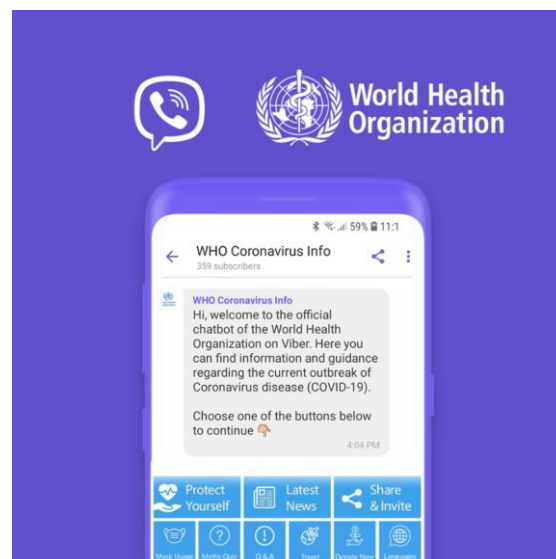
²³ Allen, J., & Rand, D. (2024, September 30). *Combating misinformation runs deeper than swatting away "fake news."* Scientific American. <https://www.scientificamerican.com/article/combating-misinformation-runs-deeper-than-swatting-away-fake-news/>

²⁴ Trachtenberg, A., & Wt. (2024, February 9). *Pov: Health misinformation is rampant on social media.* Boston University. <https://www.bu.edu/articles/2024/health-misinformation-rampant-on-social-media/>

²⁵They will also be partnering with Google for the advertisement of targeted ads on public health issues, while they will adapt the WHO health messages in language and cultures of local audiences. This is then further amplified by YouTube by being an influencer for the WHO to help reach out with the most relatable content to the target audience. Analytical data will track this effort, but the WHO will make use of online campaigns and, on occasion, change the algorithms to push critical health information in critical situations.²⁶

Partnering with WhatsApp and Facebook, WHO launched dedicated messaging services in several languages, including Arabic, English, French, Hindi, Italian, Portuguese and Spanish, to share critical guidance on COVID-19.²⁷ This easy-to-use messaging service could reach up to 2 billion people and allows WHO to get the facts directly into people's hands. The WHO also partnered with Rakuten Viber to launch a new interactive chatbot which aims to get accurate information about COVID-19 to people in multiple languages. This partnership gives WHO the potential to reach over 1 billion people in their local language directly through their mobile phones.²⁸ Moreover, the WHO and the International Telecommunication Union (ITU), with support from the United Nations Children's Fund (UNICEF), called upon all telecommunication companies worldwide to join their initiative to help unleash the power of communication technology to save lives from COVID-19 through

text messages. An estimated 3.6 billion people remained offline, most of whom live in low-income countries, where an average of just two out of every ten people are online.



WHO's interactive chatbot for accurate information
Source: WHO

Information Network for Epidemics (EPI-WIN)

Due to the high demand for timely and trustworthy information about COVID-19, WHO has established the Information Network for Epidemics (EPI-WIN) that unites technical and social media teams working closely to track and respond to misinformation, myths and rumours and provide tailored information and evidence for action. For instance, when the first confirmed case of COVID-19 was reported in Uganda, some communities in the country's north believed that

²⁵Gongane, V. U., Munot, M. V., & Anuse, A. D. (2022). *Detection and moderation of detrimental content on social media platforms: Current status and Future Directions*. Social network analysis and mining. <https://ncbi.nlm.nih.gov/pmc/articles/PMC9444091>

²⁶Covid-19, Infodemics, and the platformization of truth. (n.d.-a). <https://journals.sagepub.com/doi/10.1177/20563051211069048>

²⁷World Health Organization. (n.d.-d). *Who health alert brings covid-19 facts to billions via WhatsApp*. World Health Organization. <https://www.who.int/news-room/feature-stories/detail/who-health-alert-brings-covid-19-facts-to-billions-via-whatsapp>

²⁸World Health Organization. (n.d.-a). *1st who infodemiology conference, who infodemic management*. World Health Organization. <https://www.who.int/teams/epi-win/infodemic-management/1st-who-infodemiology-conference>

the virus was caused by evil spirits and performed rituals to “push the spirits back to the West.” Hate speech targeted at foreigners within the communities began to increase. The Ugandan Government took on a more active role in addressing the virus and disseminating accurate information from the WHO. Many precautionary measures from the WHO and the Government were translated into local languages to raise awareness among various communities, especially those in remote, hard-to-reach areas with limited access to digital information. United Nations country teams and missions on the ground used all available channels, such as radio and social media to dispel rumours and counter misinformation, while the Organization's 59 UN Information Centres also took lead in fighting misinformation in local languages.

Working with media and journalists

The United Nations Educational, Scientific and Cultural Organization (UNESCO) has published two policy briefs that assess the COVID-19 ‘disinfodemic’ of falsehoods, fabrications and misinformation. The policy briefs were supported by the International Center for Journalists (ICJ), which is assisting journalists working on the frontlines of the “disinfodemic” around the world, to ensure accurate, trustworthy and verifiable public health information reaches communities everywhere. The goals of these policy briefs were:

- Effectively contain and counter dangerous falsehoods caused by misinformation and disinformation;

- Assist UNESCO Member States to align their disinfodemic responses to international human rights standards on freedom of expression, access to information, and privacy;
- Empower citizens through media and information literacy skills;
- Support quality independent journalism to provide verifiable and reliable information.²⁹

UNESCO's International Programme for the Development of Communication (IPDC) Bureau had approved multiple initiatives across Africa, India, and the Caribbean as well. The initiatives are designed to:

- Connect 200 women journalists from Eastern Africa via an online hub that provides training materials on outbreak reporting.
- Support 25 community radio stations in Eastern and Southern Africa, which collectively reach 250,000 rural and marginalised residents.
- Support 25 community media outlets in collaboration with disaster management authorities to communicate with wider audiences, specifically tribal and marginalised groups.
- Equip 50 media professionals across nine Eastern Caribbean countries to enhance fact-checking.³⁰

²⁹New UNESCO Policy Briefs launched assessing the COVID-19 ‘Disinfodemic’. (n.d.). UNESCO.

<https://www.unesco.org/en/articles/new-unesco-policy-briefs-launched-assessing-covid-19-disinfodemic?hub=70>

³⁰United Nations. (n.d.). 5 ways the UN is fighting “infodemic” of misinformation. United Nations.

<https://www.un.org/en/un-coronavirus-communications-team/five-ways-united-nations-fighting-%E2%80%99infodemic%E2%80%99-misinformation>

Global Infodemiology Conference (June 2020)

The first Global Infodemiology Conference was organised by WHO which primarily focused on discussing strategies to mitigate the spread of misinformation during health crises. It took place from 29th June to 21st July 2020. Experts from multiple disciplines were invited to find tools and chart out agendas for improved infodemic management. The meeting was crucial in forming a global strategy against the threats of misinformation brought by the outbreak. It had 4 primary objectives:

- Understand the multidisciplinary nature of infodemic management.
- Identify current examples and tools to understand, measure and control infodemics.
- Build a public health research agenda to direct focus and investment in this emerging scientific field.
- Establish a community of practice and research.

It involved various participants ranging from UN agencies, to public health organisations and various experts in health, social studies, epidemiology, research and journalism, community engagement, ethics and other relevant disciplines engaging in discussion.³¹

The conference concluded with a call to action for global stakeholders to invest in infodemiology as

a fundamental tool to enhance public health resilience and preparedness against future health crises, as well as discussing key shortcomings of strategies used at the time for infodemic management.

Infodemic Management Framework

Infodemic Management consists of measures that are taken to manage the situation wherein there is an overload of information both false or misleading during a global catastrophic event.

As challenging as infodemics can be, the WHO believes that it can be addressed with the same principles of any communicable disease outbreak³². The organisation has been able to come up with a number of approaches together with frameworks to lessen the ill effects of misinformation and disinformation.

Key Pillars of the Infodemic Management Framework:³³

- WHO employs the social listening technique to scour the web for disinformation that is spread across social networks, the news, or online communities.
- Developing resilience towards misinformation by engaging the community through media and health literacy.
- Engaging and empowering the individual and the community: It is

³¹Public Health Informatics. Public Health Informatics - an overview | ScienceDirect Topics. (n.d.). <https://www.sciencedirect.com/topics/medicine-and-dentistry/public-health-informatics>

³²Jacobsen, K. H., Aguirre, A. A., Bailey, C. L., Baranova, A. V., Crooks, A. T., Croitoru, A., Delamater, P. L., Gupta, J., Kehn-Hall, K., Narayanan, A., Pierobon, M., Rowan, K. E., Schwebach, J. R., Seshaiyer, P., Sklarew, D. M., Stefanidis, A., & Agouris, P. (2016, March). *Lessons from the ebola outbreak: Action items for emerging infectious disease preparedness and response*. EcoHealth. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7087787/>

³³Eysenbach, G. (2020, June 29). *How to fight an Infodemic: The four pillars of Infodemic management*. Journal of medical Internet research. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7332253/>

advisable for well respected figures in the community to actively participate in countering the falsehoods of the community. Health workers, community leaders, and social media figures are involved so that such messages reach the intended audience more efficiently.

- Timely identification and reporting of misinformation: Once a lie is recognized, the WHO teams up with fact-checking bodies and various stakeholders to offer the most reliable information within the shortest time possible.
- Strengthening information environments: WHO has collaborated with Google, Twitter and Facebook in order to further strengthen the emphases of their algorithms.

The WHO is at the forefront through global infodemic management training, also promoting the study of infodemiology. It encompasses techniques in the study of the spread and determinants of information in the digital and physical environment, as well as its consequences for public health. Moreover, the organisation will conduct original research on the causes and effects of infodemics and share knowledge gleaned from those studies through webinars, workshops, and publications that will feed into global best practices-in-the-making. It is part of a larger effort to make infodemic management a science-and-evidence-driven activity.

Infodemic Management Activities by the WHO have various benefits. Firstly, there is a lesser spread of misinformation. WHO partnerships with tech companies have, in turn, resulted in increased chances of identifying fake claims and harmful content flagged or removed for removal.

The prompt release of accurate information has helped people make adequate health decisions. The emphasis by WHO on media as well as health literacy empowered people with the capability of identifying misinformation and therefore decreased the relevant effects on society.

Digital Literacy Programs

WHO calls for digital competence as one of the measures towards protecting people from fake news. The organisation is tasked with the responsibility of engaging government and educational institutions in development of public health campaigns and materials that inform people. Amidst order issuing campaigns, social media combat, or WHO directly combating misinformation, WHO has initiated various methods to develop health literacy to tackle these epidemics. Here are some core initiatives and efforts in the form of Digital Literacy Programs:

- Digital Literacy Campaigns: WHO directly partners with social media businesses including Twitter, Facebook, Youtube, and TikTok to promote valid health information and curb misinformation. Apart from fact-checking campaigns, they offer mechanisms for users to judge the validity of health information posted online.³⁴
- Digital Health Literacy toolkit: WHO has developed toolkits aimed at enhancing the health literacy levels of the public so as to increase the likelihood of fully understanding credible health information. This includes resources on countering online falsehoods, appraising information and resource

³⁴U.S. National Library of Medicine. (1970, January 1). *WHO guideline recommendations on Digital Interventions for Health System strengthening*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/books/NBK541902/>

sites as well as important health concepts.

- WHO Information Network for Epidemics (EPI-WIN): EPI-WIN delivers reliable, evidence-based messaging during health emergencies to various stakeholders that include health and non-health sectors.

CASE STUDIES

COVID-19

COVID-19, caused by the SARS-CoV-2 virus, is an infectious respiratory disease characterised by symptoms ranging from mild respiratory issues to severe pneumonia and, in some cases, it can lead to organ failure and death. First identified in Wuhan, China, in late 2019, the disease quickly escalated into a global pandemic, impacting millions worldwide and causing unprecedented health, economic, and social disruptions.³⁵ The COVID-19 pandemic brought not only physical and mental health issues but also an "infodemic" crisis.

The fast dissemination of misinformation regarding COVID-19, for example, inconceivable explanations about its origin, measures to prevent, and how to treat it, caused confusion and fear among the public. Social media platforms became places where conspiracy theories and wrong health advice were propagated. The circulation of misinformation decreased the trust of individuals in public health authorities, which in turn, resulted in vaccine side effects and disobedience to safety measures. A considerable number of people tended to doubt the efficiency of vaccines because of the contradictory information. Conflicting narratives

concerning the virus divided society and politics, which in turn, determined the compliance with the public health guidelines and the policies.³⁶

Algorithm based platforms like social media sites and search engines spread misinformation quickly by promoting sensationalistic, emotionalist high-engagement content. The rapid spread of information on an exponentially commercialised level known as "viral virality" made the dissemination of spurious explanations for what was happening with COVID-19 in terms of its emergence, symptoms and treatment widely and quickly proliferated. This was accompanied by a surge in misinformation, fake cures and rumours around conspiracy theories and vaccine hesitancy narratives fuelled further by echo-chambers where users were more probable to interact with views that supported their own opinions.

For instance, some debates on mask mandates and lockdowns became very tensed in numerous countries, even having political oppositions taking extreme sides on the situation at hand. In response to the information tsunami, the WHO launched the "Managing the COVID-19 Infodemic" initiative which was designed to give the right information and eliminate false claims about COVID-19. WHO carried out systematic communication strategies, including the #SafeHands campaign, to promote hygiene practices and raise awareness of COVID-19 symptoms and prevention. Moreover, they partnered up with social media influencers and organisations so as to gain more widespread support. WHO brought in together global research to collect the evidence about the virus and its transmission as well as effective

³⁵Cascella, M. (2023, August 18). *Features, evaluation, and treatment of coronavirus (COVID-19)*. StatPearls [Internet]. <https://www.ncbi.nlm.nih.gov/books/NBK554776/>

³⁶Collins, F. S. (2024, September 16). *Why didn't facing a common enemy bring us together?* The Atlantic. <https://www.theatlantic.com/ideas/archive/2024/09/covid-political-polarization-public-trust-vaccine/679806/>

treatments.³⁷ Their constant amendments of the guidelines, which were based on newly developed data, made it easier for medical practitioners and the public to stay informed. A survey by Pew Research Center discovered that as much as 61% of participants were worried of an increase in misleading information about COVID-19, while approximately 48% were confused about the accuracy of different information sources.

Ebola Outbreak

Ebola, caused by the Ebola virus, is a severe and often fatal illness characterised by fever, severe bleeding, organ failure, and, in many cases, leads to death if not treated early. First discovered in 1976 in Central Africa, Ebola outbreaks are highly infectious, spreading through direct contact with bodily fluids of infected individuals or contaminated surfaces.³⁸

From 2014 to 2016 there was the West African Ebola outbreak which caused over 28,600 confirmed cases and more than 11,300 deaths.³⁹ The Democratic Republic of Congo is also a region where some other smaller outbreaks have occurred. During the Ebola outbreaks, the West African outbreak in particular, misinformation relating to the transmission, treatment and prevention proliferated. Some myths such as Ebola being spread through supernatural means

or through mosquito bites spread community wide panic, and hindered individuals and households from approaching the issue in a coherent manner. Misinformation brought about the problem of increased fear and stigmatisation towards Ebola patients, which in turn led to discrimination against survivors and healthcare workers. This stigmatisation caused health directives to be ineffective and resulted in people being hesitant to consider treatment. This state of public perception remained a key challenge, especially in rural areas where backward notions often proliferate. In certain areas, misinformation was the reason for people to be rebellious against the public health interventions, that is, the vaccination and safe burial practices. A study showed that about 25% of the population in the affected regions believed that Ebola was a conspiracy created by foreign governments.⁴⁰ Communities were sceptical of the health authorities, thinking they were either misinformed or had hidden agendas, such as disruptive local cultures and practices.

³⁷Ducharme, J. (2024, September 16). *Did the pandemic break our brains?*. Time. <https://time.com/7021575/covid-pandemic-19-brain-cognition/>

³⁸World Health Organization. (n.d.-b). *Ebola virus disease*. World Health Organization. https://www.who.int/health-topics/ebola#tab=tab_1

³⁹Jacobsen, K. H., Aguirre, A. A., Bailey, C. L., Baranova, A. V., Crooks, A. T., Croitoru, A., Delamater, P. L., Gupta, J., Kehn-Hall, K., Narayanan, A., Pierobon, M., Rowan, K. E., Schwebach, J. R., Seshaiyer, P., Sklarew, D. M., Stefanidis, A., & Agouris, P. (2016, March). *Lessons from the ebola outbreak: Action items for emerging infectious disease preparedness and response*. EcoHealth. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7087787/>

⁴⁰Loukatou, S., Fakourelis, P., Papageorgiou, L., Megalooikonomou, V., Kossida, S., & Vlachakis, D. (2014). *Ebola virus epidemic: A deliberate accident?*. Journal of molecular biochemistry. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6114937/#:~:text=The%20origin%20of%20Ebola,except%20a%20few%20basic%20characteristics>



Burial of a victim of Ebola
Source: Concern Worldwide

The WHO focused on rapid outbreak response, by declaring the West African Ebola outbreak a Public Health Emergency of International Concern (PHEIC) in August 2014. They organised international clinics, sent medical teams to the countries that were affected, and provided the local health workers with resources and training. They also prioritised public communication campaigns, where the WHO targeted communication strategies to combat misinformation. They collaborated with the local leaders and organisations to inform people who have wrong ideas that Ebola viral disease is highly contagious and can be prevented by vaccination. The "Ebola: How to Protect Yourself" campaign was one of the initiatives that were launched to educate the public. Research and vaccine development was also a key contribution for treatments by the WHO. The vaccine rVSV-ZEBOV was developed and used during the outbreak. The rVSV-ZEBOV vaccine was used in the DRC outbreaks between 2018 and 2019, during which more than 500,000 people were vaccinated.⁴¹

To conclude, the Ebola experience provided some lessons especially in handling crises in relation to dissemination of information and the perception that the public has concerning certain diseases. In addition, WHO responded proactively and supported sharing accurate health information

and vaccines significantly contributed to managing the effects of Ebola. The information obtained from previous Ebola outbreaks can be of great value for handling future outbreaks of communicable diseases.

CONCLUSION

All in all, the spread of false information has become a threat to public health, which in the digital age stands transformed with the ongoing infodemic. In this regard the WHO has developed a comprehensive approach that encompasses various methodologies to deal with the threat of misinformation. The WHO has developed the Global Infodemic Management Framework which is considered as a revolutionary model that merges evidence oriented research with practical solutions to combat false narratives developed on diverse online platforms. As such, working hand in hand with authorities, tech companies, academia, and public sector actors on all levels allows WHO to improve information effectiveness to contribute to global good.

Other frameworks and past actions also include running basic and applied research on infodemiology as developing infodemic awareness tools that integrate big data, social media content, and psychology in the mission of combating insufficiencies. With this, WHO goes an extra benchmark ahead by considering the culture bound where regional determinants matter. This allows the organisation to stay two steps ahead and respond proactively rather than having a reactive approach. Media education and general public information campaigns will enable media and digital literacy. This multifaceted approach highlights WHO's dedication to strengthen the health information ecosystem that enables other related industries in the long run.

⁴¹World Health Organization. (n.d.-c). *Ebola virus vaccines*. World Health Organization.
<https://www.who.int/groups/global-advisory-committee-on-vaccine-safety/topics/ebola-virus-vaccines>